An Application for the Separation of Nuts and Bolts using Image Processing and Artificial Intelligence

Amit A.Kale
Dept. of Electronics Engg.
S.B.Jain Institute of Technology, Management & Research, Nagpur

Dhiraj S Bhojane
Dept. of Electronics Engg.
Datta Meghe Institute of Technology & Research, Nagpur

Dr. Prashil M Junghare
Dept. of ETC
S. B. Jain Institute of Technology, Management & Research, Nagpur

***Abstract*— Particularly in the automotive mechanical industries which are the manufacturer of nuts and bolts. There is need to design a system that recognised the nuts and bolts. The main aim of this study to develop a system to recognise the difference in nuts and bolts for their separation. For the processing of this system , the input is the normalised image of nuts and bolts with its size and dimentions and it can be used for the detection. With this idea, A system can be designed using the digital image processing and artificial intellegnce. The system can be design using different algorithms and these all algorithms are integrated using MATLAB software. The image processing algorithm will develop using wavelet transform which can be used to crop the images and this images can be used further as a suitable input for processing. The task of the recognition system developer is to devise a system that takes the data from a sensor or a high performance camera and assign a class for example nuts or bolts. This image is then applied for feature extraction by using principle component analysis. For the detection of an object Artificial intelegence system used. The resultant from feature extraction is given to Artificial intellegence for detection of object and then applied for the required application.**

***Keywords*: digital image processing, MATLAB, Artificial intellegence, principle component analysis.**

I. INTRODUCTION

Humans can recognize any object by the natural logical thinking process by human perception of identification. But machines are not having the perception of identification for recognizing of an object as it is far away from the human recognition system. So research is to be develop for increasing the recognition efficiency of the machine

A challenging task is to authenticate determination of the nuts and bolts during its production. So the determination parameters are design depending on the size quality of the nuts and bolts. But these parameters are changes or vary from machine to machine. Machines are made up of various parts to extract the image from an object to solve the same task. To solve the problem, a branch of engineering and technology is used called as computer vision. Computer vision is a technique use to extract the information from the various images based on automated approach useful in the automobile industries. In computer vision, a high performance camera performs the function of human eye which minimizes the human biological system in visual system. Camera is used to capture the targeted images for this system. These images are then stored in the memory of computer which are further used for process of recognition. For the sorting of nuts and bolts on a conveyer belt in mechanical industries, the wavelet transform technique is used. Wavelet transform is implemented using MATLAB for image processing and detection. The soft computing technique of artificial intelligence is used to train the data sets of nuts and bolts.

Artificial neural network is technique which works like a human brain to retain the data as it is first identified or trained. The main objective of this project to study the various feature extraction technique and then finalized the one of the technique for this project for getting faster and perfect result.

II. SOFTWARE FRAMEWORK

In this paper we are discussing in two processes these are as Principle component analysis and wavelet transform.

The flow chart of this process is shown below

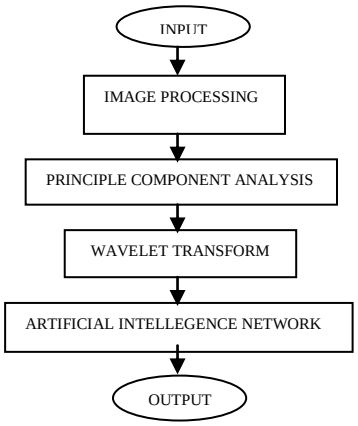

Figure1: Software Framework

The input is obtained from the high performance camera which is used to capture the images of nuts and bolts on the conveyer belt. These images are then process by using digital image processing tool for the better enhancement of the image.

III. PROPOSED SYSTEM

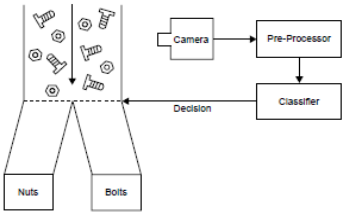

Figure2: Proposed System

In the above mentioned diagram, the nuts and bolts are transfer on the conveyer belt after the production has been done. This paper is used for the application that separates the nuts and bolts and gives the output as the separated nuts and bolts. For the separation of the product three basic components are used these are camera, Preprocessor and classifier.

In this application a high performance and high resolution camera is used to capture the images of nuts and bolts. These images are then given to preprocessor for the further analysis. This process is called image acquisition. The preprocessor is used to remove the noise, deblurring that are present in the image so that noise free image can be given for the feature extraction. This is called image enhancement. However the noise free image is used for the feature extraction. For the feature extraction, the principal component analysis is used. The principal component analysis is used to calculate the eigen values for feature extraction. After extracting the data it is given for the classification and identification to the classifier. Artificial neural network is for the classification. In this paper, back propagation algorithm is for the classification and identification. After identification it is given to decision making for the separation of the nuts and bolts.

IV. RESULTS

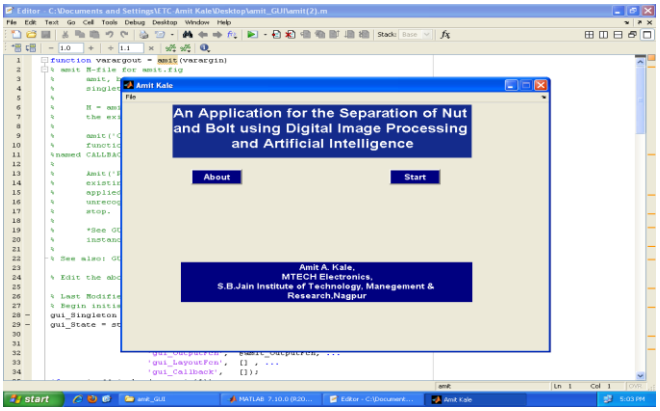

Figure3: GUI Interface Diagram

When we run the project, the first GUI graphical interface indicates the name of the concern operation and the person who have made this GUI.

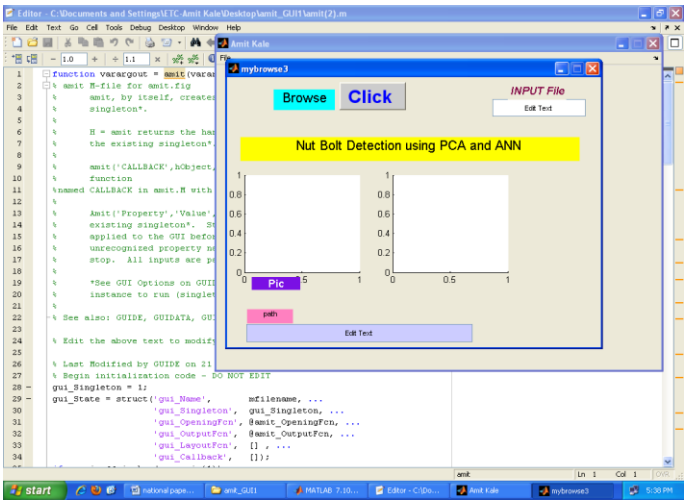

Figure4: Detection of parts using PCA &ANN

Once we click on next button, figure 4 is to be appeared.

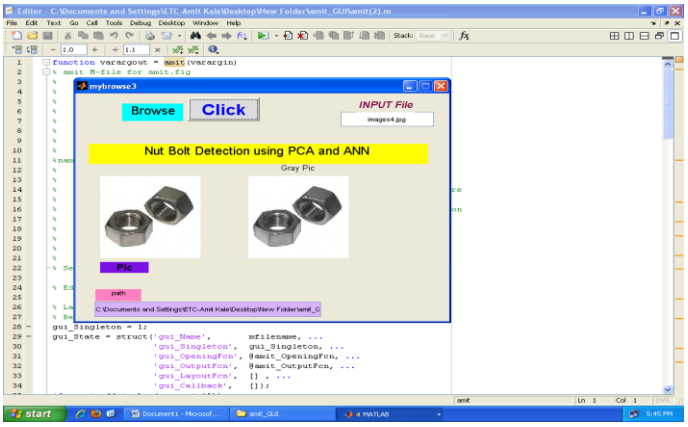

Figure5: Image detection & Capteruziation

When we click on browse, the image which we have already captured by camera will be appeared in first block and conversion of colour image in gray scale is appeared on second block.

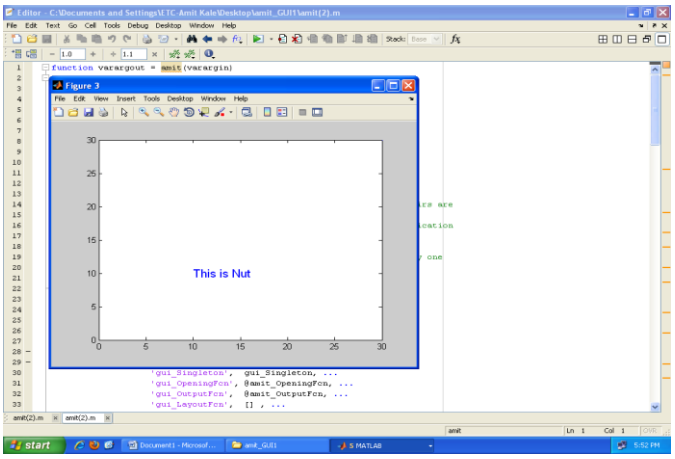

Figure6: Detection of Part

This GUI indicates which part is detected which is appeared in fig. no. 5.

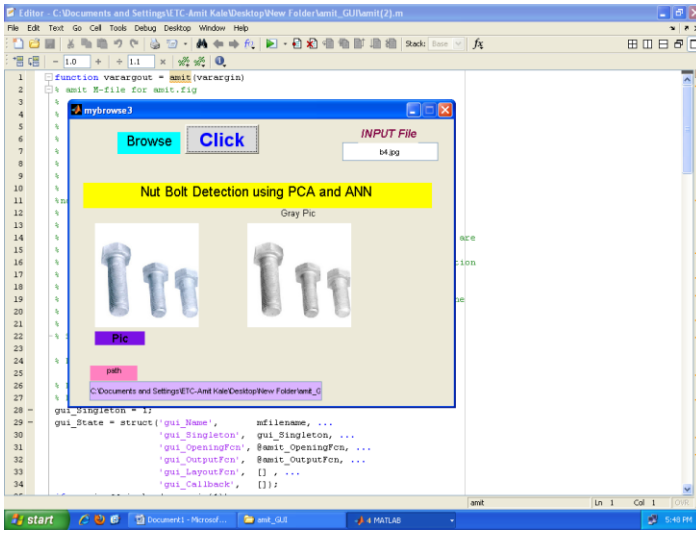

Figure7: Capitalization of Image

This GUI captures the image of bolt by camera and its conversion in gray scale.

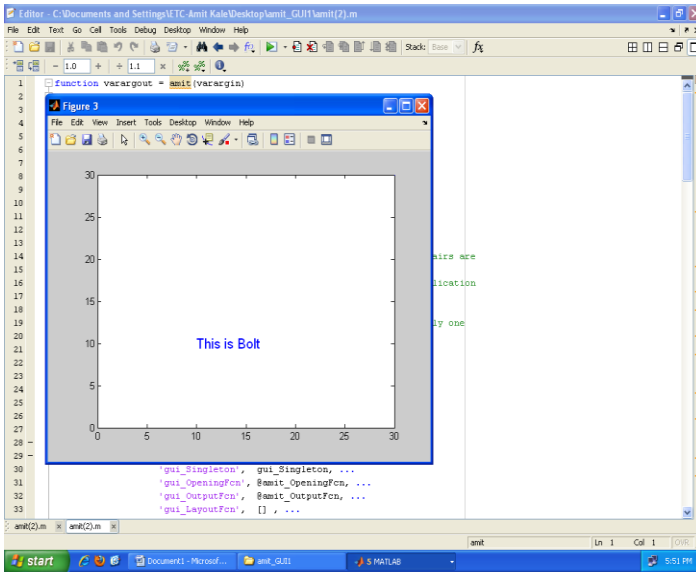

Figure8: Analysis of parts

This GUI indicates which part is detected which is appeared in fig. no. 7.

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